



National  
Qualifications  
2024

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# **2024 Music Technology**

## **Higher**

### **Question Paper Finalised Marking Instructions**

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## General marking principles for Higher Music Technology

*Always apply these general principles. Use them in conjunction with the detailed marking instructions, which identify the key features required in candidates' responses.*

- (a) Marking should always be positive. This means that, for each candidate response, marks are accumulated for the demonstration of relevant skills, knowledge and understanding: they are not deducted from a maximum on the basis of errors or omissions.
- (b) Do not accept any answer that is not in the marking instructions. Use professional judgement when candidates' responses do not exactly match marking instructions but carry the same meaning.
- (c) Accept inaccurate spelling of English or Italian terms, as long as the meaning is clear.
- (d) Where a question asks for a description, explanation or justification, do not award marks for simple one or two word answers.

## Marking instructions for each question

| Question |     |  | Expected response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Max mark                   | Additional guidance                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------|-----|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.       | (a) |  | Jazz funk<br>Wah-wah                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2                          | 1 mark for each.<br>Do not accept other answers here.                                                                                                                                                                                                                                                                                                                                                                  |
|          | (b) |  | <ul style="list-style-type: none"> <li>• jazz instrumentation</li> <li>• melodic and/or complex bass lines</li> <li>• ostinato and/or riff</li> <li>• prominent bass line</li> <li>• syncopation</li> <li>• the excerpt has a (jazz) head</li> <li>• use of (Hammond) organ</li> <li>• use of electric guitar</li> <li>• use of electric piano</li> <li>• use of synthesizer.</li> </ul>                                                                                                                                                | 2                          | 1 mark for each.<br>Also accept any answer that infers jazz instrumentation.<br><br>Also accept (funk) head.<br>Also accept Rhodes piano.                                                                                                                                                                                                                                                                              |
| 2.       | (a) |  | <p>R'n'B</p> <p><b>Music features</b></p> <ul style="list-style-type: none"> <li>• brass or horn section</li> <li>• use of electric guitar*</li> <li>• use of electric piano*</li> <li>• ostinato or riff</li> <li>• soulful vocals</li> <li>• syncopation</li> <li>• use of synthesisers*</li> <li>• vocal harmonies</li> </ul> <p><b>Technological features</b></p> <ul style="list-style-type: none"> <li>• bass prominent in mix</li> <li>• loops</li> <li>• programmed drum beats</li> <li>• samples</li> <li>• wah wah</li> </ul> | <p>1</p> <p>1</p> <p>1</p> | <p>1 mark for correct genre.</p> <p>1 mark for a reason related to a musical feature.</p> <p>Also accept any instrument typically found within brass or horn section.</p> <p>*Also accept as technological features but only credit once.</p> <p>Also accept melismatic vocals.</p> <p>Also accept backing vocals.</p> <p>1 mark for a reason related to a technological feature.</p> <p>Also accept drum machine.</p> |
|          | (b) |  | Tremolo                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 1                          | Except no other answer.                                                                                                                                                                                                                                                                                                                                                                                                |

| Question |     |      | Expected response                                                                                                                                                                                   | Max mark | Additional guidance                                                                                                                                                             |
|----------|-----|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3.       | (a) | (i)  | The gain on the overhead right channel is set too high.                                                                                                                                             | 1        | 1 mark for a description of the fault. Also accept there is audible distortion, clipping or peaking in the overhead right channel.                                              |
|          |     | (ii) | Adjust or lower the gain or trim.                                                                                                                                                                   | 1        | 1 mark for a detailed description of a matching correction.<br>Do not accept turn the overhead right channel's volume or level down.                                            |
| 3.       | (b) | (i)  | The threshold of a noise gate is set too high.                                                                                                                                                      | 1        | 1 mark for a description of the fault. Also accept a noise gate is set incorrectly.                                                                                             |
|          |     | (ii) | Adjust or lower the threshold.                                                                                                                                                                      | 1        | 1 mark for a detailed description of a matching correction.<br>Do not accept increase in threshold.<br>Also accept any answer that infers a reduction in noise gate settings    |
| 4.       | (a) |      | <b>Option 3 is the correct answer.</b><br>The rate is set to slow and the mix is set to max.                                                                                                        | 1        | The mark should only be awarded if answer <b>one</b> has been identified and a valid justification has been given.<br>Also accept rate is set to min and the mix is set to max. |
|          | (b) |      | Depth<br>Rate<br>Shape                                                                                                                                                                              | 2        | Also accept intensity.<br>Also accept speed.<br>Also accept, wave, form or waveform.<br>Do not accept wet, dry or mix.                                                          |
| 5.       | (a) |      | Sampler                                                                                                                                                                                             | 1        | Do not accept other answers here.                                                                                                                                               |
|          | (b) |      | Drum sequencer and reverb                                                                                                                                                                           | 1        | Do not accept other answers here.                                                                                                                                               |
|          | (c) |      | Sabrina Setlur could have: <ul style="list-style-type: none"> <li>significantly changed the audio</li> <li>obtained permission to the audio</li> <li>given a writing credit to Kraftwerk</li> </ul> | 1        | Accept any one answer.<br>Do not accept other answers here.                                                                                                                     |

| Question |     |  | Expected response                                                                                                                                          | Max mark | Additional guidance                                                                                                                                    |
|----------|-----|--|------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6.       | (a) |  | Ostinato<br>Soul                                                                                                                                           | 2        | 1 mark for each.<br>Do not accept other answers here.                                                                                                  |
|          | (b) |  | A matched or stereo pair of cardioid condensers in XY or ORTF or NOS.<br><br><b>OR</b><br>Stereo pair of omni-directional condensers in AB or spaced pair. | 1        | 1 mark for array.<br><br>Also accept spaced pair of figure of 8 microphones.<br>Also accept Faulkner array, Decca Tree and Blumlein (coincident pair). |
|          |     |  | Microphones should be placed at a metre or more in front of or above the string quartet.                                                                   | 1        | 1 mark for placement.<br>Also accept a metre from the string quartet.                                                                                  |
|          |     |  | To achieve a good balance of direct sound and room ambience.<br><br><b>OR</b><br>To ensure good stereo image.                                              | 1        | 1 mark for justification.<br><br>Also accept anything that infers good stereo balance.                                                                 |
|          | (c) |  | Whole tone scale                                                                                                                                           | 1        | Do not accept other answers here.                                                                                                                      |

| Question       |                                                                                                                                                                                                                                                                                      | Expected response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Max mark      | Additional guidance                                                                                                                                                                                                                                                                  |                |                                                                                                                    |               |                                                                                                                                                                                                     |                |                                                                                                                                                                                                                   |               |                                                                                                                                                                                                                             |                |                                                                                   |   |                                                                                                                                                                                                                                                                                                                                                  |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|--------------------------------------------------------------------------------------------------------------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-----------------------------------------------------------------------------------|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7.             | (a)                                                                                                                                                                                                                                                                                  | <p>Candidates are asked to select a genre from the list provided. No marks are awarded for this.</p> <p>For the genre selected marks are awarded for identifying:</p> <ul style="list-style-type: none"><li>• a key attribute(s) (1 mark)</li><li>• a key technology. (1 mark)</li></ul> <p>Examples of key innovators, attributes and technologies could include:</p> <p>Genre: Indie</p> <table><tr><td>Key attribute</td><td><ul style="list-style-type: none"><li>• Broken chords</li><li>• Electric guitar*</li><li>• Live sounding recording</li><li>• Lo-fi recordings</li><li>• Melancholic, raw, rough or unpolished vocal style</li><li>• Ostinato</li><li>• Picked guitar style</li><li>• Riff.</li></ul></td></tr><tr><td>Key technology</td><td><ul style="list-style-type: none"><li>• Chorus</li><li>• Delay</li><li>• Distortion</li><li>• Overdrive.</li></ul></td></tr></table> <p>Genre: Electroacoustic</p> <table><tr><td>Key attribute</td><td><ul style="list-style-type: none"><li>• Atonal</li><li>• Heavily processed acoustic sounds</li><li>• Irregular time signatures</li><li>• Sounds created or processed by electronic means.</li></ul></td></tr><tr><td>Key technology</td><td><ul style="list-style-type: none"><li>• Delay</li><li>• Loops</li><li>• Reverb</li><li>• Samplers</li><li>• Sequencers</li><li>• Synthesiser</li><li>• Tape manipulation</li><li>• Time domain effects.</li></ul></td></tr></table> <p>Genre: Punk</p> <table><tr><td>Key attribute</td><td><ul style="list-style-type: none"><li>• Aggressive, raw, rough or unpolished vocal style</li><li>• Electric guitar*</li><li>• Little post production</li><li>• Political lyrics</li><li>• Simple chord structure.</li></ul></td></tr><tr><td>Key technology</td><td><ul style="list-style-type: none"><li>• Distortion</li><li>• Overdrive.</li></ul></td></tr></table> | Key attribute | <ul style="list-style-type: none"><li>• Broken chords</li><li>• Electric guitar*</li><li>• Live sounding recording</li><li>• Lo-fi recordings</li><li>• Melancholic, raw, rough or unpolished vocal style</li><li>• Ostinato</li><li>• Picked guitar style</li><li>• Riff.</li></ul> | Key technology | <ul style="list-style-type: none"><li>• Chorus</li><li>• Delay</li><li>• Distortion</li><li>• Overdrive.</li></ul> | Key attribute | <ul style="list-style-type: none"><li>• Atonal</li><li>• Heavily processed acoustic sounds</li><li>• Irregular time signatures</li><li>• Sounds created or processed by electronic means.</li></ul> | Key technology | <ul style="list-style-type: none"><li>• Delay</li><li>• Loops</li><li>• Reverb</li><li>• Samplers</li><li>• Sequencers</li><li>• Synthesiser</li><li>• Tape manipulation</li><li>• Time domain effects.</li></ul> | Key attribute | <ul style="list-style-type: none"><li>• Aggressive, raw, rough or unpolished vocal style</li><li>• Electric guitar*</li><li>• Little post production</li><li>• Political lyrics</li><li>• Simple chord structure.</li></ul> | Key technology | <ul style="list-style-type: none"><li>• Distortion</li><li>• Overdrive.</li></ul> | 2 | <p>Candidates are asked to select a genre from the list provided. No marks are awarded for this.</p> <p>For the genre selected marks are awarded for identifying:</p> <ul style="list-style-type: none"><li>• a key attribute (1 mark)</li><li>• a key technology. (1 mark)</li></ul> <p>*Also accept as a key feature but only credit once.</p> |
| Key attribute  | <ul style="list-style-type: none"><li>• Broken chords</li><li>• Electric guitar*</li><li>• Live sounding recording</li><li>• Lo-fi recordings</li><li>• Melancholic, raw, rough or unpolished vocal style</li><li>• Ostinato</li><li>• Picked guitar style</li><li>• Riff.</li></ul> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |               |                                                                                                                                                                                                                                                                                      |                |                                                                                                                    |               |                                                                                                                                                                                                     |                |                                                                                                                                                                                                                   |               |                                                                                                                                                                                                                             |                |                                                                                   |   |                                                                                                                                                                                                                                                                                                                                                  |
| Key technology | <ul style="list-style-type: none"><li>• Chorus</li><li>• Delay</li><li>• Distortion</li><li>• Overdrive.</li></ul>                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |               |                                                                                                                                                                                                                                                                                      |                |                                                                                                                    |               |                                                                                                                                                                                                     |                |                                                                                                                                                                                                                   |               |                                                                                                                                                                                                                             |                |                                                                                   |   |                                                                                                                                                                                                                                                                                                                                                  |
| Key attribute  | <ul style="list-style-type: none"><li>• Atonal</li><li>• Heavily processed acoustic sounds</li><li>• Irregular time signatures</li><li>• Sounds created or processed by electronic means.</li></ul>                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |               |                                                                                                                                                                                                                                                                                      |                |                                                                                                                    |               |                                                                                                                                                                                                     |                |                                                                                                                                                                                                                   |               |                                                                                                                                                                                                                             |                |                                                                                   |   |                                                                                                                                                                                                                                                                                                                                                  |
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| Key technology | <ul style="list-style-type: none"><li>• Distortion</li><li>• Overdrive.</li></ul>                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |               |                                                                                                                                                                                                                                                                                      |                |                                                                                                                    |               |                                                                                                                                                                                                     |                |                                                                                                                                                                                                                   |               |                                                                                                                                                                                                                             |                |                                                                                   |   |                                                                                                                                                                                                                                                                                                                                                  |

| Question |     |   | Expected response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Max mark | Additional guidance                                                                                                                                                                                                                                                                       |
|----------|-----|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | (b) |   | <p><b>Gramophone</b></p> <p><b>Example of key innovator:</b></p> <ul style="list-style-type: none"> <li>Emile Berliner</li> </ul> <p><b>Example of their influence on the development of this technology:</b></p> <ul style="list-style-type: none"> <li>Berliner developed a turntable that could play hard, shellac records.</li> </ul> <p><b>OR</b></p> <ul style="list-style-type: none"> <li>Shellac records were easier to duplicate than wax cylinders.</li> </ul> <p><b>Radio</b></p> <p><b>Example of key innovator:</b></p> <ul style="list-style-type: none"> <li>Guglielmo Marconi</li> </ul> <p><b>Example of their influence on the development of this technology:</b></p> <ul style="list-style-type: none"> <li>Marconi invented a transmitter and receiver which used radio waves to carry a signal.</li> </ul> <p><b>Bass Guitar</b></p> <p><b>Example of key innovator:</b></p> <ul style="list-style-type: none"> <li>Leo Fender</li> </ul> <p><b>Example of their influence on the development of this technology:</b></p> <ul style="list-style-type: none"> <li>Fender released the Precision bass guitar.</li> </ul> <p><b>OR</b></p> <ul style="list-style-type: none"> <li>Fender created one of the first commercially successful bass guitars.</li> </ul> | 2        | <p>Candidates are asked to select a technology from the list provided. No marks are awarded for this.</p> <ul style="list-style-type: none"> <li>Example of a key innovator. (1 mark)</li> <li>Example of the key innovator's development on the selected technology. (1 mark)</li> </ul> |
| 8.       |     |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          | 1 mark for each correct answer.                                                                                                                                                                                                                                                           |
|          |     | 1 | Delay Panning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1<br>1   | Also accept echo.<br>Also accept autopan.                                                                                                                                                                                                                                                 |
|          |     | 2 | Cymbal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 1        | Also accept crash (cymbal).                                                                                                                                                                                                                                                               |
|          |     | 3 | Minor                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1        | Do not accept other answers here.                                                                                                                                                                                                                                                         |
|          |     | 4 | Major                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1        | Do not accept other answers here.                                                                                                                                                                                                                                                         |
|          |     | 5 | Pitch shift                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 1        | Also accept harmoniser.                                                                                                                                                                                                                                                                   |
|          |     | 6 | Release                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 1        | Do not accept other answers here.                                                                                                                                                                                                                                                         |
| 9.       |     |   | <ul style="list-style-type: none"> <li>Vocals sung through a vocoder</li> <li>Piano plays arpeggios</li> <li>Glockenspiel panned right</li> <li>Inverted pedal on string synth</li> <li>Synth with portamento.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 5        | <p>1 mark for each correct feature.</p> <p>Features can be listed in any order.</p> <p>Do not accept other answers here.</p>                                                                                                                                                              |

[END OF MARKING INSTRUCTIONS]